$\forall$ func. $f: A \to B$,

$\forall a_1, a_2 \in A$, $a_1 = a_2 \Rightarrow f(a_1) = f(a_2)$

$\exists$

f is inj. <= ...

Proof: (of $\begin{cases} f \text{ is surj.} \\ g \text{ is surj.} \end{cases} \Rightarrow g \circ f$ is surj.

f, g are surj.

$f: A \to B$

$g: B \to C$

let $z \in C$

g is surj. $\Rightarrow \exists y \in B$, s.t. $g(y) = z$

f is surj. $\Rightarrow \exists x$ ... $f(x) = y$

then $g \circ f(x) = z \Rightarrow g \circ f$ is surj.

$$\begin{pmatrix} f: A \to B \\ g: B \to C \end{pmatrix}$$

$\forall$ func. g, f, g∘f is inj. $\Rightarrow$ f is inj.)

→ ①

Proof:

$f: A \to B$

$g: B \to C$

Let g∘f be inj.

② then f is inj. ← ①

Let $a_1, a_2 \in A$

Assume that $f(a_1), f(a_2) \in B$

→ ②

g is a function $\Rightarrow g(f(a_1)) = g(f(a_2))$

$\Rightarrow g \circ f(a_1) = g \circ f(a_2)$

g∘f is inj. $\Rightarrow$ f is inj.

$$\begin{pmatrix} f: A \to B \\ g: B \to C \end{pmatrix}$$

$\forall$ func. g, f, $g \circ f$ is surj. $\Rightarrow$ g is surj.

Proof:

Let $\forall z \in C$

$g \circ f$ is surj.

$\Rightarrow \exists x \in A$, s.t. $g \circ f(x) = z$

Denote $y = f(x) \in B$,

then $g(y) = \cdots = z$

~~($\because \exists y \in B$ and $y = f(x)$, s.t. $g(y) = z$)~~

①

$f(a) = b_1$,

$g(b_1) = g(b_2) = c$ ~~X~~

$g \circ f(a) = c$ ~~bij.~~ ②

②

$g \circ f$ is inj.

g is not inj.

$b_1 \neq b_2$ but $g(b_1) = g(b_2) = c$

$g \circ f$ is surj.

f is not surj.

$f(A) = \{b_1\} \subsetneq B$

(~~f~~ inverse)

~~(1)~~

let $f: A \to B$ be bij.

$\exists\, g: B \to A$

and $\begin{cases} \forall a \in A,\ g(f(a)) = a \\ \forall b \in B,\ f(g(b)) = b \end{cases}$

Then $g = f^{-1}$

and $f = g^{-1}$

Thm.: Let A, B be finite sets,

s.t. $|A| = m$ and $|B| = n$, where $m, n \in \mathbb{N}$

(~~Statements~~ :-)

~~(3)~~ ① $|A| \le B$

② $\exists f: A \to B$ is inj.

③ $\exists g: B \to A$ is surj.

Proof:

(① ⇒ ②):

Asume that $m \le n$

$A = \{a_1, \cdots, a_m\}$ where $a_1, \cdots, a_m$ are distinct

$B = \{b_1, \cdots, b_n\}$ where $b_1, \cdots, b_n$ are distinct

Define $f: A \to B$,

s.t. $\forall i \in [1, m]$, $f(a_i) = b_i$

Then $\forall a_i \neq a_j$, $i \neq j$, $b_i \neq b_j$, so $f(a_i) \neq f(a_j)$

$\Rightarrow f$ is inj.

(② ⇒ ③):

Let $f: A \to B$ be inj.

W.L.O.G., asume that $\forall i \in [1, m]$, $b_i = f(a_i)$

Define $g: B \to A$,

s.t. $g(b) = \begin{cases} a_i & \text{if } \exists i \in [1, m], \text{ s.t. } b = b_i \\ a_1 & \text{if } \forall i \in [1, m], b \neq b_i \end{cases}$

So $\forall i \in [1, m]$, $a_i = g(b_i)$

$\Rightarrow g$ is surj.

Corollary:

$\forall$ sets A, B,

$|A| = |B| \Rightarrow \exists f: A \to B$ is bij.

Proof:

($\Rightarrow$):

Assume that $|A| = m = n = |B|$

By [thm. ① $\Rightarrow$ ②],

$\exists f: A \to B$ is inj.

Then $|f(A)| = m = |A|$

② $|B| = m \Rightarrow f$ is bij.

($\Leftarrow$):

2

$f: A \to B$,

$U \subseteq A$, $V \subseteq B$

$\left(f(U) = \{ f(x) : x \in U \} \right.$

$\left. f^{-1}(V) = \{ a : f(a) \in V \} \right)$

$\forall X, Y \subseteq B$,

$f^{-1}(X \cup Y) = f^{-1}(X) \cup f^{-1}(Y)$

$f^{-1}(X \cap Y) = f^{-1}(X) \cap f^{-1}(Y)$

$\forall C, D \subseteq A$,

$f(C \cap D) \subseteq f(C) \cap f(D)$

Proof:

($\subseteq$)

Let $y \in f(C \cap D)$

then $\exists x \in C \cap D$, s.t. $y = f(x)$

$x \in C \cap D \Rightarrow \begin{cases} x \in C \Rightarrow y \in f(C) \\ x \in D \Rightarrow y \in f(D) \end{cases}$

$\Rightarrow y \in f(C) \cap f(D)$

$\Rightarrow f(C \cap D) \subseteq f(C) \cap f(D)$

($\supseteq$ if f is inj).

Let $\forall y \in f(C) \cap f(D)$

$$\Rightarrow \begin{cases} y \in f(C) \Rightarrow \exists x_1 \in C, \text{ s.t. } y = f(x_1) \\ y \in f(D) \Rightarrow \exists x_2 \in D, \text{ s.t. } y = f(x_2) \end{cases}$$

$$\Rightarrow f(x_1) = f(x_2)$$

f is inj. $\Rightarrow x_1 = x_2$

Let $x = x_1 = x_2$

Then $\begin{cases} x \in C \\ x \in D \end{cases} \Rightarrow \underline{x \in C \cap D}$

$y = f(x_1) \Rightarrow \underline{y = f(x)} \Rightarrow y \in f(C \cap D)$

Induction

$\forall n \in \mathbb{N}$, statements $P(n)$,

($\rightarrow P(n): \sum_{k=1}^{n} k = \frac{(n+1)\cdot n}{2}$

(e.g.:)

(i) $\forall n \in \mathbb{N}, (P(n) \Rightarrow P(n+1)) \wedge P(1)$

Principle of induction

Let $(P(n))_{n \in \mathbb{N}}$ be a seq. of natural numbers

$$\begin{cases} \forall n \in \mathbb{N}, P(n) \Rightarrow P(n+1) \\ P(1) \text{ is true} \end{cases} \Rightarrow \forall n \in \mathbb{N}, P(n) \text{ is true}$$

E.g.: 证明 $P(n): \sum_{k=1}^{n} k = \frac{(n+1)\cdot n}{2}$

Base case:

Let $n=1$:

$\sum_{k=1}^{1} k = 1$, $\frac{(1+1)\times 1}{2} = 1$, $1=1$

$P(1)$ is true

Induction step

Let $n \geq 1$

Assume that $P(n)$ is true

(Goal! 证明 $P(n+1): \sum_{k=1}^{n+1} k = \frac{(n+2)(n+1)}{2}$ is true)

LHS: ——

RHS: ···

(Remete $S_n = \sum_{k=1}^{n} k$

Then $2S_n = 1+2+\cdots+n$

$= n+n-1+\cdots+1$

$= n\cdot(n+1)$

$\Rightarrow S_n = \frac{n\cdot(n+1)}{2}$)

E.g.: 证 $\forall n \in \mathbb{N}$, $P(n): \sum_{k=1}^{n}(3k+2) = \frac{3n^2+7n}{2}$

Base case:

let $n=1$

Induction step:

let $n \geq 1$ / Assume that $P(n)$ is true

(Goal: ---) ($= \frac{3n^2+7n}{2}$)

$\sum_{k=1}^{n+1}(3k+2) = [\sum_{k=1}^{n}(3k+2)] + [3(n+1)+2]$

LHS: --- (用 $P(n)$)

RHS: --- (用 $P(n)$)

E.g.: 证: $\forall n \in \mathbb{N}$, $12 \mid 19^n - 7^n$

$\Leftrightarrow 19^n \equiv 7^n \pmod{12}$

$\Leftrightarrow 19^n - 7^n \equiv 0 \pmod{12}$



3

(前至後)

induction step: let $n \geq 1$

Assume $P(n)$ ($\checkmark$) ($12 \mid 19^n - 7^n$) $\rightarrow$ ($= k \cdot 12$)

(WTS: $P(n+1): 12 \mid 19^{n+1} - 7^{n+1}$)

$19^{n+1} - 7^{n+1} = (12+7) \cdot 19^n - 7^{n+1}$

$= 12 \cdot 19^n + 7 \cdot (19^n - 7^n)$

$= 12 \cdot (19^n + 7k)$

By ind. ---- ①

By lemma, ---- ②

Hence ---- ($P(n+1)$ $\checkmark$)

---- ③

E.g.: $\forall n \in \mathbb{N}$, $\forall$ real $x_1, \cdots, x_n \in [0,1]$,

$$P(n): \prod_{k=1}^{n} (1-x_k) \geq 1 - \sum_{k=1}^{n} x_k$$

Proof:

$n=1$: $1-x_1 \geq 1-x_1$ ✓

$n \geq 1$: Assume $P(n)$ ✓

(⇒ ① $n+1$) Let real $x_1, \cdots, x_{n+1} \in [0,1]$

$$\prod_{k=1}^{n+1} (1-x_k) = \left[\prod_{k=1}^{n} (1-x_k)\right] \cdot (1-x_{n+1})$$

$$\geq \left(1 - \sum_{k=1}^{n} x_k\right) \cdot \text{——}$$

$$= 1 + x_{n+1}\left(\sum_{k=1}^{n} x_k\right) - x_{n+1} - \sum_{k=1}^{n} x_k$$

$$= 1 + \text{——} - \sum_{k=1}^{n+1} x_k$$

$$\geq$$

≥ 0

(⇐ ① 一開)

$\forall i \in [1,n]$, $x_i \geq y_i$

Prove: $P(n): \sum_{i=1}^{n} x_i \geq \sum_{i=1}^{n} y_i$

$n=1$: $x_1 \geq y_1$ ✓

$n \geq 1$: Assume $P(n)$ ✓

(⇒ ① $n+1$)

(⇐ ① 一開)

Well-ordered

Def.: set $A \subseteq \mathbb{R}$ is well-o.

$\Leftarrow \forall B \subseteq A$ and $B \neq \emptyset$,

$\exists b = \min B$

($b \in B$, s.t. $\forall y \in B$, $b \leq y$

$(0,1)$ x well-o.

$[0,1]$ x well-o. ~~=~~

Axiom: $\mathbb{N}$ is well-o.

Thm.: ($P(n)$ is a seq. of statements, where $n \in \mathbb{N}$

~~induction~~) $P(1)$ ✓

$\forall n \in \mathbb{N}$, $(P(n) \Rightarrow P(n+1))$ ✓

$\Rightarrow \forall n \in \mathbb{N}$, $P(n)$ ✓

Proof: Assume that the thm. is X

↳ $\Rightarrow \exists n_0 \in \mathbb{N}$, s.t. $P(n_0)$ X

Define $S = \{ n \in \mathbb{N} : P(n) \text{ X} \} \subseteq \mathbb{N}$

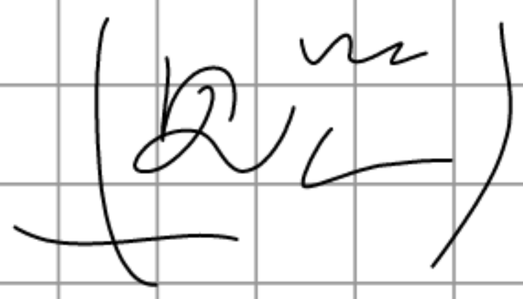
We have $n_0 \in S \Rightarrow S \neq \emptyset$

(axiom → $\mathbb{N}$ is well-o. $\Rightarrow \exists m = \min S$

$S \subseteq \mathbb{N}$

$m \in S \Rightarrow P(m)$ X





Method of infinite descent

E.g. Assume $\sqrt{2} \in \mathbb{Q}$

$\Rightarrow \exists x, y \in \mathbb{N}$, s.t. $\sqrt{2} = \frac{x}{y}$

Then $x^2 = 2y^2$

Consider $S = \{ x \in \mathbb{N} : \exists y \in \mathbb{N}, \text{ s.t. } x^2 = 2y^2 \}$

$x \in S \Rightarrow S \neq \emptyset$

By well-o. of $\mathbb{N}$, ($\leftarrow$ $S \subseteq \mathbb{N}$)

$\exists m = \min S$

$\Rightarrow m \in S$

$\Rightarrow \exists n \in \mathbb{N}$, s.t. $m^2 = 2n^2$

$2n^2 = m^2$ is even

m is even

$\Rightarrow \exists k \in \mathbb{N}$,

s.t. $m = 2k$